William M. Rohren

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Education

Texas A&M University - GPA : 3.126 — **Expected Dec 2026**
BSME: *Bachelor of Science in Mechanical Engineering* — *College Station, TX*

Work Experience

Rapid Prototyping Studio (RPS) — **Jan 2024 — Present**
Student Technician — *College Station, TX*

- Coordinated the use of >$75,000 of equipment to help process 1,000+ manufacturing requests per semester.
- Supervised shop operations, overseeing $10,000+ of material consumption & ensuring student safety.
- Trained over 1,000 students on prototyping tooling and created training material using Wiki.js and Markdown.
- Repaired $12,000 Torchmate CNC plasma cutter and authored a detailed SOP for its safe handling & usage.
- Kickstarted construction of a 5'x5'x5' large-format FDM 3D printer for creation of Aero-Shells for SAE formula teams.

Industrial Engineering Dept. — **May 2025 – Aug 2025**
Undergraduate Research Assistant — *College Station, TX*

- Coordinated and led the design of a custom autoclavable binder-jet CoreXY 3D printer for use in the microbiology dept.
- Operated an $80,000 ExOne Binder-jet 3D printer to facilitate compaction studies for Ph.D students with various materials including tungsten, titanium, steel, aluminum-ceramics, silicon carbide, and more.
- Hosted several team meetings and workshops to teach peers about CAD/CAM/PCB-design/etc.

Organizations

SAE Aero Design Team — **May 2025 – Present**
Structures & Material Science (SMS) Engineer — *College Station, TX*

- Designing structural members of plane components used in competition, as well as using FEA to simulate deflection/stress under load, and verify positive margin.
- Iterated between topology optimization, classical FEA, and manual design to translate optimized geometries into manufacturable balsa and basswood structures.
- Collaborated with the Aerodynamics, Stability, & Controls (ASC) subteam in order to translate their aerodynamic shell into components that can be manufactured out of balsa and bass wood members.

Aggies Create — **Aug 2022 — May 2023**
Member — *College Station, TX*

- Designed and built a smart filament storage box for a startup company in Austin (Accelerate 3D) (Fall '22 - Spring '23)
- Researched and prototyped a regenerative braking system for integration in SRF-Gen4 race car hybrid powertrain (Fall '23)

Projects

Atmega32U4 Development Board — **Aug 2024 — Jan 2025**

- Created a compact embedded dev board from PCB design on Altium to physical prototype PCBs with JLCPCB.
- Implemented low-power modes for optional battery operation.

Arduino Alarm Clock — **Jan 2023 — Jun 2024**

- Designed and built a custom alarm clock using Arduino, teaching myself digital circuit theory and C++ programming.
- Simulated in TinkerCAD, prototyped on breadboards, and designed/assembled a custom PCB in Altium.

Skills

CAD & CAM
SolidWorks, Fusion 360, Inventor

FEA Software
Ansys Mechanical, Femap, Topology Optimization

Programming & Embedded Sys.
MATLAB, Python, C++, ATmega, ESP32

PCB Design
Altium, KiCAD, THT/SMD Soldering

Machining & Fabrication
3/4/5 Axis CNC Mill, Lathe

3D Printing
FDM, MSLA, SLS, BJT / Repairs